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Recell startup

Supervised Learning, Foundations

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January 3, 2024

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Executive Summary

Based on the first machine learning model, after checking for all assumptions of linear regression, the following independent variables have the most influence over the pricing of refurbished used devices.

- used_phone_age
- used_tablet_age
- used_phone_price
- used_tablet_price
- used_phone_battery
- used_tablet_battery
- used_phone_screen
- used_tablet_screen
- used_phone_os
- used_tablet_os
- used_phone_memory
- used_tablet_memory
- used_phone_storage
- used_tablet_storage
- used_phone_color
- used_tablet_color
- used_phone_brand
- used_tablet_brand
- used_phone_model
- used_tablet_model
- used_phone_year
- used_tablet_year
- used_phone_weight
- used_tablet_weight
- used_phone_height
- used_tablet_height
- used_phone_width
- used_tablet_width
- used_phone_depth
- used_tablet_depth
- used_phone_material
- used_tablet_material
- used_phone_finish
- used_tablet_finish
- used_phone_condition
- used_tablet_condition
- used_phone_location
- used_tablet_location
- used_phone_date
- used_tablet_date
- used_phone_time
- used_tablet_time
- used_phone_user
- used_tablet_user
- used_phone_company
- used_tablet_company
- used_phone_department
- used_tablet_department
- used_phone_division
- used_tablet_division
- used_phone_group
- used_tablet_group
- used_phone_team
- used_tablet_team
- used_phone_project
- used_tablet_project
- used_phone_task
- used_tablet_task
- used_phone_role
- used_tablet_role
- used_phone_title
- used_tablet_title
- used_phone_desc
- used_tablet_desc
- used_phone_note
- used_tablet_note
- used_phone_comment
- used_tablet_comment
- used_phone_review
- used_tablet_review
- used_phone_rating
- used_tablet_rating
- used_phone_feedback
- used_tablet_feedback
- used_phone_suggestion
- used_tablet_suggestion
- used_phone_idea
- used_tablet_idea
- used_phone_proposal
- used_tablet_proposal
- used_phone_plan
- used_tablet_plan
- used_phone_strategy
- used_tablet_strategy
- used_phone_tactic
- used_tablet_tactic
- used_phone_technique
- used_tablet_technique
- used_phone_skill
- used_tablet_skill
- used_phone_capability
- used_tablet_capability
- used_phone_proficiency
- used_tablet_proficiency
- used_phone_experience
- used_tablet_experience
- used_phone_expertise
- used_tablet_expertise
- used_phone_knowledge
- used_tablet_knowledge
- used_phone_insight
- used_tablet_insight
- used_phone_understanding
- used_tablet_understanding
- used_phone_comprehension
- used_tablet_comprehension
- used_phone_grasp
- used_tablet_grasp
- used_phone_apprehension
- used_tablet_apprehension
- used_phone_perception
- used_tablet_perception
- used_phone_sensation
- used_tablet_sensation
- used_phone_feeling
- used_tablet_feeling
- used_phone_emotion
- used_tablet_emotion
- used_phone_mood
- used_tablet_mood
- used_phone_manner
- used_tablet_manner
- used_phone_demeanor
- used_tablet_demeanor
- used_phone_disposition
- used_tablet_disposition
- used_phone_temperament
- used_tablet_temperament
- used_phone_nature
- used_tablet_nature
- used_phone_character
- used_tablet_character
- used_phone_trait
- used_tablet_trait
- used_phone_quality
- used_tablet_quality
- used_phone_attribute
- used_tablet_attribute
- used_phone_feature
- used_tablet_feature
- used_phone_characteristic
- used_tablet_characteristic
- used_phone_property
- used_tablet_property
- used_phone_quality
- used_tablet_quality
- used_phone_attribute
- used_tablet_attribute
- used_phone_feature
- used_tablet_feature
- used_phone_characteristic
- used_tablet_characteristic
- used_phone_property
- used_tablet_property

Another recommendation is to hone in on specialized markets of customers interested in devices with a certain capacity, i.e. large screen for entertainment, or large battery capacity for use on the road, in order to further optimize sales.

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Business Problem Overview and Solution Approach

Business Problem Overview

The used and refurbished device market has grown considerably over the past decade. This growth can be attributed to an uptick in demand for used phones and tablets that offer considerable savings to both consumers and businesses compared with new models. Recell, a startup aiming to tap the potential in this market, has identified the need to develop a dynamic pricing strategy for used and refurbished devices.

Solution approach and methodology

In this report I used a Machine Learning based solution approach to develop such a dynamic pricing strategy. I analyzed the dataset provided by Recell using univariate and bivariate analysis and preprocessed the data where needed. Next I built a linear regression model to predict the price of a used phone/tablet, and identified factors that significantly influence it.

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EDA Results - Univariate analysis

- In the univariate analysis of this dataset, outliers feature heavily in the distribution of data for certain columns or features of the used / refurbished device. See the Appendix for detailed code, graphs and plots. The reasons for outliers can for the majority be explained by the wide variety of features, and the wide range within those features in cell phone and tablet devices in general.

Note: You can use more than one slide if needed

[Link to Appendix slides and data background context](#)

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EDA Results - Univariate analysis

Resolution of the rear camera in megapixels - example of outlier effect

Observation:



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